

HIPPO: An Iterative Reparametrization Method for Identification and Calibration of Dynamic Bioreactor Models of Complex Processes

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Supporting Information

ABSTRACT: Unstructured, dynamic bioreactor models of complex processes usually possess many nonidentifiable, insensitive, or statistically nonsignificant parameters. However, an exhaustive search to find a reduced set of identifiable parameters is computationally demanding. We developed a heuristic iterative procedure for parameter optimization (HIPPO), generic and free of symbolic mathematical manipulations, to help bioprocess engineers to find and estimate a set of significant parameters, thus obtaining reliable models. This methodology includes local sensitivity, significance and identifiability analysis, metaheuristic optimization, and fitting performance assessments. We tested this method with two bioreactor models: a microalgal fed-batch bioreactor (MFB) 12-parameter model and a solid substrate fermentation (SSF) 14-parameter model. Applying this procedure, we were able to find a MFB reparametrization with 4 fitting parameters after 479 iterations and a SSF reparametrization with 6 fitting parameters after 918 iterations. In both cases, all fitting parameters were statistically significant and accurately estimated and models had a better fit than the originally fitted 12- and 14-parameter models. The heuristic was programmed using MATLAB and is freely available to the research community (<http://www.systemsbiology.cl/tools/>).

INTRODUCTION

The main purpose of any model's fitting procedure is to yield a model that can be used with confidence to explain and predict the behavior of a given system.^{1–4} Otherwise, the model would just fit the data obtained in a particular experimental condition but would have poor accuracy in predicting the behavior of the system in other circumstances or scenarios. Usually, models with an excessive number of adjustable parameters can fit data very well, but confidence intervals of some estimated parameters are large and model predictions are biased.⁵

Traditionally, local gradient-based algorithms are used for parameter estimation, which work well if the model is smooth and continuous. However, many useful process models are nonsmooth and present discontinuities. Hence, global metaheuristic search methods are being increasingly used instead of traditional algorithms for parameter estimation.^{6,7}

After parameter estimation, and for testing the quality of a given model fitting exercise, Jaqaman and Danuser³ have proposed applying a series of procedures, named pre/post-regression analysis, consisting of parameter sensitivity (how variations on parameters affect model predictions),⁸ parameter significance (if a parameter is significantly different from zero), and parameter identifiability⁹ (correlation degree between parameters). If results are discouraging after pre/post-regression analysis (for instance, if there is a high correlation between pairs of parameters or wide confidence intervals), the model has to be modified by reestimating parameters on new scenarios or fixing some less sensitive parameters.

The decision on which parameters should be fixed is relatively arbitrary when more than one parameter presents any of the aforementioned situations. Reestimation of parameters (or reparametrization) should be therefore done iteratively, fixing and excluding from the next estimation procedure one or more

parameters at a time and exploring several combinations. If the model is nonlinear and the operating range varies widely, the result of the mentioned analysis is dependent on local operating points (i.e., the estimated parameter values). To overcome this local analysis limitation, parameters ought to be estimated with a global optimization method to ensure that parameters' significance and identifiability are assessed in a proper region of the optimization search space.¹⁰ Additionally, when the parameter set is large, an exhaustive search gets computationally intractable. Therefore, previous knowledge and intuition should be applied to reduce the search space of possible parameter combination sets; in practice this is best achieved by analyzing real process data in the first place.

Iterative model fitting methods for reparametrization (also named sometimes parameter identifiability analysis or parameter selection) have been presented before. Jacquez and Greif⁹ proposed a comprehensive agenda of experimental design, parameter identifiability analysis, and parameter estimation. However, their analysis is based on linearization of process dynamic equations, which is not accurate enough when Monod-like growth rates and curvature effects (unavoidable in unstructured bioreactor models¹¹) appear. More recently, Balsa-Canto et al.² have proposed an iterative identification procedure that uses structural identifiability analysis, global ranking of parameters to select the most relevant ones, global optimization methods, and optimal experimental design. The problem here is that extensive model-specific symbolic manipulations are necessary, and therefore the approach cannot

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be used generically in any model without considerable effort from the modeler. Several approaches have also been presented for parameter selection based on local sensitivity analysis,¹² including heuristic and optimization-based methods. In all of these approaches, nonetheless, the parameters are studied only around one point, without iterative parameter reestimation. Moreover, extensive computational power is often required. Finally, set-by-set selection¹³ has been recently proposed as a procedure that alternates parameter estimation and identifiability analysis, with promising results.

In this work we present HIPPO (heuristic iterative procedure for parameter optimization), a simple approach for fitting parameters in dynamic models. At each iteration step, the model parameters are reestimated using a scatter search code⁷ to increase the chance of getting global optimal estimates, and pre/post-regression procedures are applied. Based on these results, one or more parameters are fixed so smaller sets of fitting parameters are generated for the following iterations. The procedure uses heuristic rules to reduce the number of parameter combinations to be analyzed. This methodology was recently adapted by our group to analyze a dynamic genome-scale metabolic model of *Saccharomyces cerevisiae*.¹⁴ However, here we show its applicability to any given kinetic model, and we also make available the software tool to the academic community (<http://www.systemsbiology.cl/tools/>).

We have applied HIPPO to find a reduced set of identifiable model parameters for two dynamic bioreactor models taken from literature: a microalgal fed-batch bioreactor model (MFB) developed by Surisetty et al.¹⁵ and a solid substrate fermentation (SSF) model developed by Gelmi et al.¹⁶ Both models included a large number of fitting parameters, and although those authors have proposed methods to improve model identifiability (ad-hoc procedures using nonlinear functions in the MFB case¹⁷ and a one-step reparametrization based on a pre/post-regression analysis to the original model in the SSF case¹), the method developed in this work is shown to have significant advantages over those approaches and is expected to be useful for engineers modeling similar processes.

METHODS

Iterative Heuristic Procedure. To achieve a model in which all estimated parameters are significant, we use an iterative heuristic procedure for parameter optimization (HIPPO), comprising global metaheuristic parameter estimation and pre/post-regression diagnostics. The procedure starts by estimating all model parameters followed by regression diagnostics; then if a parameter presents problems, it will be removed from the estimation set and its value fixed to its first estimate; afterwards new parameter estimation and regression diagnostics are performed on the reduced set of parameters. This iteration proceeds until all estimated parameters are significant. In order to select candidate parameters to be fixed, the following rules are applied:

(I) Identifiability: Pearson correlation coefficient with any other parameter larger than 0.95 (this threshold value can be changed in HIPPO according to the researcher needs). If this is the case the interactions between the corresponding pair of parameters is so large that separate numerical estimates of them are impractical.^{9,18}

(II) Sensitivity: an almost null influence on the model, i.e., if the normalized sensitivity score is below 0.01 (see Pre/Post-Regression Diagnostics).

(III) Significance: a value not significantly different from zero, i.e., confidence coefficient smaller than 2 (from the significance test outcome; see Pre/Post-Regression Diagnostics).

If more than one parameter satisfies at least one of the previously mentioned rules in a given iteration, each fixed parameter generates an additional exploratory branch (i.e., a reduced set of parameters to be reestimated). Hence, a growing exploratory tree is generated because identifiability, sensitivity, and/or significance problems may appear in different branches. Since the iterative procedure can easily get intractable as many parameter combinations could be explored, trimming criteria (TC) are included. Our aim is not to find the best possible parameter set, which will require excessive iterations. Instead, we expect to find with moderate effort a reasonably good model fit. Therefore, to reduce the size of the exploratory tree, a branch is no longer explored if it satisfies one of the following trimming criteria:

(TCI) The reparametrization proposed has all parameters fixed for one of the model's differential equations. This is impractical because if the experimental setting changes, a calibration to the variable's corresponding new data will not be possible. Also, it is a requirement for model identifiability to have at least one parameter affecting each of the model's state variables.³ The implementation of this criterion is model-specific and is indicated in the Supporting Information for each case-study model.

(TCII) One of the pre/post-regression analyses of the procedure generates inconsistent results (for instance, relative sensitivity scores over 10). This result is a clear indication that the fitted parameter choice does not lie in a stable region of the combinatorial space of possible reparametrizations and therefore is a calibration with few possibilities of being predictive.

(TCIII) One of the pre/post-regression analyses of the procedure resulted in an increase of identifiability/sensitivity/significance problems in the branch, by a given factor. This factor is model-specific and was determined to be 1.2 for models with a low number of correlations, such as the MFB model, and 1.5 for models with a high number of correlations, such as the SSF model (see details in the Supporting Information).

A flowchart of the iterative procedure is depicted in Figure 1. At a given iteration, m parameters must be reestimated. First TCI is checked; if it is true, the m -parameter set is discarded from the list. Otherwise, the m parameters are estimated and identifiability, sensitivity, and significance tests are carried out. Then TCII is applied to verify that the pre/post-regression diagnostics gave consistent output; if there are inconsistencies, this m -parameter set is discarded from the list; otherwise the pre/post-regression results are analyzed. If no problems are present in any of the m estimated parameters, the corrected Akaike information criterion (AIC_c) and confidence intervals for the estimated parameters are computed and saved. However, if there are still problems in one of the m parameters, TCIII is evaluated: if this criterion is met the m -parameter set is discarded from the list; otherwise, new sets of $m - 1$ parameters are generated. All of these new parameters will be analyzed further using the same iterative scheme.

It is worth noticing that the only iteration that does not work as depicted in Figure 1 is the first one, in which none of the three trimming criteria is applied. Finally, and as expected, the procedure will iterate until all branches of the tree are explored.

After the iterative procedure finishes, all saved parameter combinations are compared and the parameter set with the lowest AIC and smaller mean confidence coefficient is chosen (for this purpose, a score was computed that adds both normalized terms, as detailed in the Supporting Information).

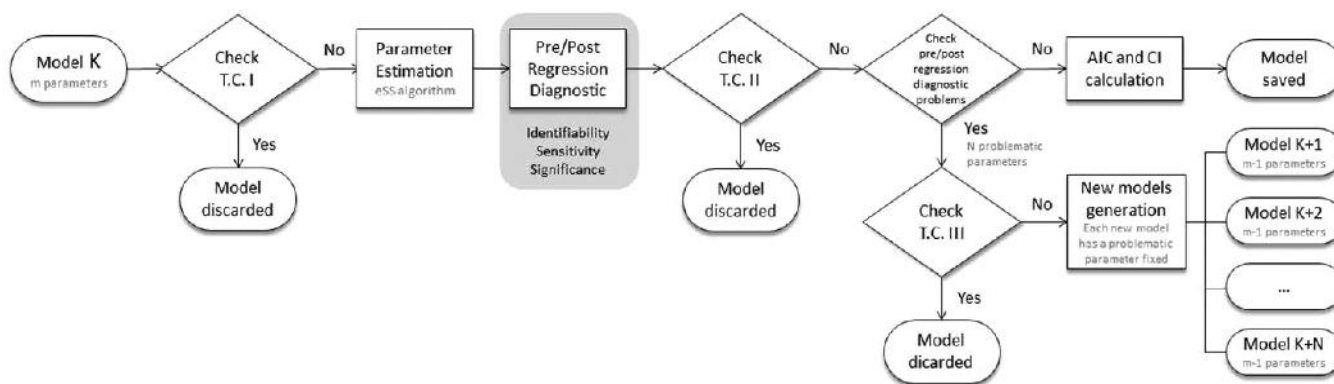


Figure 1. Flowchart for a given iteration of the reparametrization heuristic procedure. The iteration starts with a model with m parameters and ends in one of three options: the model is discarded, the model is saved, or several new combinations of the same model with $m - 1$ parameters are generated, in which case the iteration starts over for each new combination of parameters.

This improved solution should fit additional data well with a reduced set of parameters.

Parameter Estimation. Parameter estimation was formulated as a nonlinear programming problem with differential and algebraic constraints, where the objective function was the sum of weighted square errors between averaged experimental data and simulation values. To test the iterative procedure better, we used different weights for each objective function: for the MFB model weights were defined according to the averaged standard deviation of each variable measurement (eq 1), and for the SSF model the maximum averaged measured value of each variable (eq 2) was used. Both functions homogenized the scale of the different variables, but in different ways: in eq 1 data with less dispersion get a higher weight, whereas in eq 2 all data get equivalent weights. These weights were chosen for each problem after preliminary tests (data not shown).

$$F_1 = \min \sum_{l=1}^N \sum_{i=1}^n \left(\frac{x_{il}^{\text{model}} - \bar{x}_{il}^{\text{exp}}}{\bar{\sigma}_i} \right)^2 \quad (1)$$

$$F_2 = \min \sum_{l=1}^N \sum_{i=1}^n \left(\frac{x_{il}^{\text{model}} - \bar{x}_{il}^{\text{exp}}}{\max_l(\bar{x}_{il}^{\text{exp}})} \right)^2 \quad (2)$$

Here, N is the number of measurements (the same for each measured variable), n the number of measured variables, $\bar{x}_{il}^{\text{exp}}$ the average measured value of data point l of variable i , x_{il}^{model} the model prediction for the aforementioned value, and $\bar{\sigma}_i$ the average standard deviation for each measurable variable.

For parameter estimation, scatter search methods SSm⁷ and eSS¹⁹ were used in our procedure. SSm is a population-based metaheuristic global optimization algorithm that has been successfully applied to solve nonlinear programming problems with differential and algebraic constraints of complex bioprocesses.^{1,20,21} The SSm code includes effective diversification and intensification methods that keep a diverse and high-quality reference set of solutions. At each step of the optimization process, the algorithm creates new solutions from this reference set using linear or generalized linear combinations. The reference set is then updated with some of the best and more diverse of these new solutions, which prevents the algorithm from getting trapped in a local optimum. Consequently, it is expected that the obtained solution is close to the global optimum (although this cannot be guaranteed in metaheuristic optimization). A complete description of the algorithm can be found in Egea et al.⁷ The

second scatter search method is eSS, an enhanced version of scatter search that has been optimized for nonlinear programs including noise and discontinuities; further details can be found in Egea and Balsa-Canto.¹⁹

To obtain a good fit in a reasonable computational time, it is important to tune the scatter search and integration options according to each dynamic model's characteristics. For the MFB model, the best scatter search method in preliminary tests (less CPU time and lower objective function values) was eSS, in accordance to some model features (e.g., discontinuous carbon and nitrogen feed; see the Supporting Information for more details). The best options found were as follows: 1,000 maximum iterations, local optimization solver *fmincon*, and *ode23t* integration solver with relative and absolute tolerance of 1×10^{-3} and 1×10^{-6} , respectively. With these options, optimization took around 7 min to find parameter estimates in a Windows PC with a 3.3 GHz AMD FX 6100 six-core processor. For the SSF model (less complex in structure), the best choices were as follows: SSm (scatter search method), 10,000 maximum iterations; *n2fb* local optimization solver;²² and *ode23* ODE solver with 1×10^{-4} relative tolerance and 1×10^{-7} absolute tolerance. In this case, an average calibration took 5 min in the same computer mentioned previously.

It should be noted that because the first optimization of the procedure (with no fixed parameters) determined the values assigned in all subsequent reparametrizations (as mentioned in the previous section), the maximum number of iterations for the first parameter estimation (for both models) was set 1 order of magnitude higher (i.e., 10,000 for the MFB model and 100,000 for the SSF model). Finally, and also for both models, all parameter estimations (as well as all mathematical computations) were performed using the software package MATLAB²³ (version R2010a).

Pre/Post-Regression Diagnostics. The regression diagnosis at each cycle of the iterative procedure included a sensitivity analysis of each parameter with respect to each state variable, as well as an identifiability analysis of model parameters, calculations of AIC_c, and a parameter significance analysis.³

A sensitivity analysis establishes which parameters should be estimated accurately, given their high impact on the model's output, and also which parameters could be eliminated from the estimation set or estimated with low accuracy, given their low impact on the model's output. To facilitate comparison, in this work we used relative sensitivities:

$$\bar{G}_{kj}(t, \theta_j) = \frac{\theta_j}{x_k} \frac{dx_k(t)}{d\theta_j} \quad (3)$$

A normalized score of the sensitivity for each parameter over each variable was calculated as proposed by Hao et al.²⁴ Thus, if a parameter had no sensitivity score over 0.01, the parameter was fixed. A nonnormalized version of the sensitivities was stored in a sensitivity matrix ($G(t)$), whose j th column represents the sensitivity of parameter θ_j on all state variables, spanning all experimental times. The function *corrcoef* of MATLAB was used to compute the matrix of correlation coefficients (C) between each column of all sensitivity matrices, using the standard Pearson correlation formula. This matrix was used to identify those parameters that were strongly correlated; in this work we have used $|C_{ij}| > 0.95$ to select highly correlated parameters. A high correlation between any pair of parameters implies that several combinations of paired estimated values yield the same objective function in a fitting procedure. We have chosen a threshold of 0.95 because smaller values generated more branches in the algorithm, resulting in larger computational times (data not shown); nonetheless that threshold value can be changed by the user.

The values of AIC_c were calculated as follows.^{25,26}

$$AIC_c = N \ln\left(\frac{F}{N}\right) + 2(m+1) + 2\frac{(m+1)(m+2)}{N-m-2} \quad (4)$$

where F is the sum of square errors between experimental and simulated values, m is the number of model parameters, and N is the number of measurements per variable. The confidence interval (CI) for each parameter was calculated as

$$CI_j = [\hat{\theta}_j \pm 1.96\sigma_j] \quad (5)$$

where $\hat{\theta}_j$ is the estimated value of the parameter j and σ_j is the standard deviation of the parameter calculated using the Fisher information matrix (FIM, eq 8). In addition, coefficients of confidence (CC) were calculated to determine statistical significance for each parameter,

$$CC_j = \frac{\Delta(CI_j)}{\hat{\theta}_j} \quad (6)$$

Here $\Delta(CI_j)$ is the width of the CI. Statistically significant parameters (i.e., different from zero) are those whose respective CI do not contain the zero value, and therefore CC should be smaller than 2. Finally, estimated parameter variances for CI calculations were the diagonal values of the inverse of FIM:^{27,28}

$$\sigma_j^2 = FIM_{jj}^{-1} \quad (7)$$

FIM was obtained as follows,²⁸

$$FIM = G_1^T Q_1 G_1 \quad (8)$$

where Q_1 is the weighting matrix built as the inverse measurement error covariance matrix assuming white and uncorrelated noise and G_1 is the sensitivity matrix for the measurements.

RESULTS AND DISCUSSION

In this section we compare the fittings obtained with the iterative heuristic procedure with the one reported by Surisetty et al.¹⁷ for the MFB model, and the one reported by Sacher et al.⁴ for the SSF model (temperature of 25 °C and water activity of 0.999). We also present the results of the pre/post-regression analysis before and after the procedure and additional considerations regarding the proposed methodology.

Fitting Performance. Microalgal Fed-Batch Bioreactor Model. For the MFB model, the procedure took 479 iterations,

which is considerably less than all $\sum_{i=0}^{12} \binom{12}{i} = 4,095$ possible

combinations of fixed parameters (see Figure S1a in the Supporting Information). From the mentioned 479 iterations, 110 parameter sets had no identifiability, sensitivity, or significance problems (as described in Methods). Table S1 in the Supporting Information shows a comparison between all mentioned reparametrizations. Among these, the best set (according to a score which considers AIC_c and mean CC calculations; for more details see the Supporting Information) had the following 4 fitting parameters: μ_m , ρ_m , k_m , and π_m . Consequently, the fixed parameters were as follows: q_{min} , k_q , k_s , S_0 , Y_{SX} , Y_{SP} , k_p , and Y_{qX} . The values of the fitted and fixed parameters, in comparison to those of the Surisetty et al.¹⁷ solution and the original 12-parameter estimation with eSS, are shown in Table 1.

Figure 2 compares experimental data with predictions obtained with 3 models: the original 12-parameter model reported by Surisetty et al.¹⁷ as well as the 4-parameter and 12-parameter models fitted with eSS. It can be observed that the eSS fitting obtained with the 4-parameter and the 12-parameter models were similar, and more importantly, both calibrations showed better fittings than the one proposed by Surisetty et al.,¹⁷ especially in the measurable biomass and carbon source concentrations.

For further comparison, objective functions (F_1 , eq 1) for the three calibrations discussed previously are shown in Table 2. The highest F_1 is from the calibration obtained in Surisetty et al.¹⁷ and the lowest is from the calibration achieved with the 4-parameter model, thus verifying that a calibration with eSS improves fitting quality and further improvement can be achieved with our iterative procedure. Although it can be seen counterintuitive that a model with fewer estimated parameters better fits the data, it should be considered that we are dealing with a complex optimization problem in which, due to nonlinearity and unidentifiability issues, a global optimum can never be guaranteed especially when the number of parameters is too large. One of the advantages of our heuristic consists of identifying a set of significant parameters and performing the estimation only with those parameters. This is crucial, because the risks of performing estimation with a set of nonsignificant parameters is that the optimization algorithms would miss the optimal parameter values due to the implicit correlations among them, thus attaining suboptimal objective functions. Our methodology identifies those highly correlated parameters and chooses a subset of them. This allows for a more efficient exploration of the parameter space.

Solid Substrate Fermentation Model. In the case of the SSF model, after 918 iterations the procedure yielded 116 sets of parameter estimates which were identifiable, significant, and sensitive. Figure S1b in the Supporting Information displays the reduction of combinatorial space of fixed parameters from

exploring every combination ($\sum_{i=0}^{14} \binom{14}{i} = 16,383$), and

Supporting Information Table S2 shows all problem-free sets that the procedure found. The best option (using the same metric mentioned before) from the mentioned group was a model with 6 fitting parameters: μ_m , k_d , k_U , m_{CO_2} , Y_{NX} , and β_m . The following parameters were fixed in the procedure: m_{O_2} , k_p , k_N , Y_{CO_2X} , Y_{O_2X} , k_i , Y_{SX} , and m_S .

Table 1. Estimated and Fixed Parameters for the MFB Model^a

Parameters	Surisetty et al. (2010) model			eSS 12-parameter model			Optimal 4-parameter model		
	Value	CI	CC	Value	CI	CC	Value	CI	CC
μ_m [1/h]	3.20E-01	$\pm 5.19\text{E-}06$	3.24E-05	1.99E-01	$\pm 8.78\text{E-}06$	8.84E-05	2.02E-01	$\pm 6.77\text{E-}07$	6.70E-06
$q_{\min}$ [g/g]	1.67E-04	$\pm 2.53\text{E-}06$	3.04E-02	8.28E-04	$\pm 1.51\text{E-}05$	3.65E-02	8.28E-04	-	-
k_q [g/g]	2.45E-01	$\pm 3.61\text{E-}06$	2.95E-05	2.27E-01	$\pm 2.57\text{E-}05$	2.26E-04	2.27E-01	-	-
ρ_m [1/h]	1.22E-02	$\pm 3.43\text{E-}06$	5.63E-04	1.29E-02	$\pm 1.95\text{E-}05$	3.02E-03	1.25E-02	$\pm 8.50\text{E-}07$	1.36E-04
k_s [g/mL]	1.00E-03	$\pm 3.75\text{E-}06$	7.50E-03	1.32E-03	$\pm 1.51\text{E-}05$	2.29E-02	1.32E-03	-	-
s_0 [g/mL]	7.40E-05	$\pm 2.38\text{E-}06$	6.43E-02	7.52E-05	$\pm 2.07\text{E-}07$	5.50E-03	7.52E-05	-	-
Y_{SX} [g/g]	6.67E-01	$\pm 7.56\text{E-}06$	2.27E-05	6.11E-01	$\pm 2.33\text{E-}05$	7.61E-05	6.11E-01	-	-
k_m [1/h]	1.39E-02	$\pm 5.41\text{E-}06$	7.78E-04	1.27E-02	$\pm 1.64\text{E-}05$	2.58E-03	1.33E-02	$\pm 9.51\text{E-}07$	1.43E-04
Y_{SP} [g/g]	4.39E-01	$\pm 3.54\text{E-}07$	1.61E-06	4.40E-01	$\pm 2.46\text{E-}05$	1.12E-04	4.40E-01	-	-
π_m [1/h]	1.64E-02	$\pm 3.05\text{E-}06$	3.72E-04	1.85E-02	$\pm 9.66\text{E-}06$	1.04E-03	1.70E-02	$\pm 7.08\text{E-}07$	8.34E-05
k_π [g/mL]	1.79E-03	$\pm 3.27\text{E-}06$	3.65E-03	1.20E-03	$\pm 1.22\text{E-}05$	2.02E-02	1.20E-03	-	-
Y_{qX} [g/g]	1.36E+01	$\pm 8.69\text{E-}05$	1.28E-05	1.51E+01	$\pm 3.96\text{E-}04$	5.26E-05	1.51E+01	-	-

^aSurisetty model is from ref 17. The estimation accuracy and significance are shown through 95% confidence intervals (CI) and coefficient of confidence (CC), respectively. The fixed parameters appear in black shaded cells. Note: 3.20E-01 represents, for example, 3.20×10^{-1} .

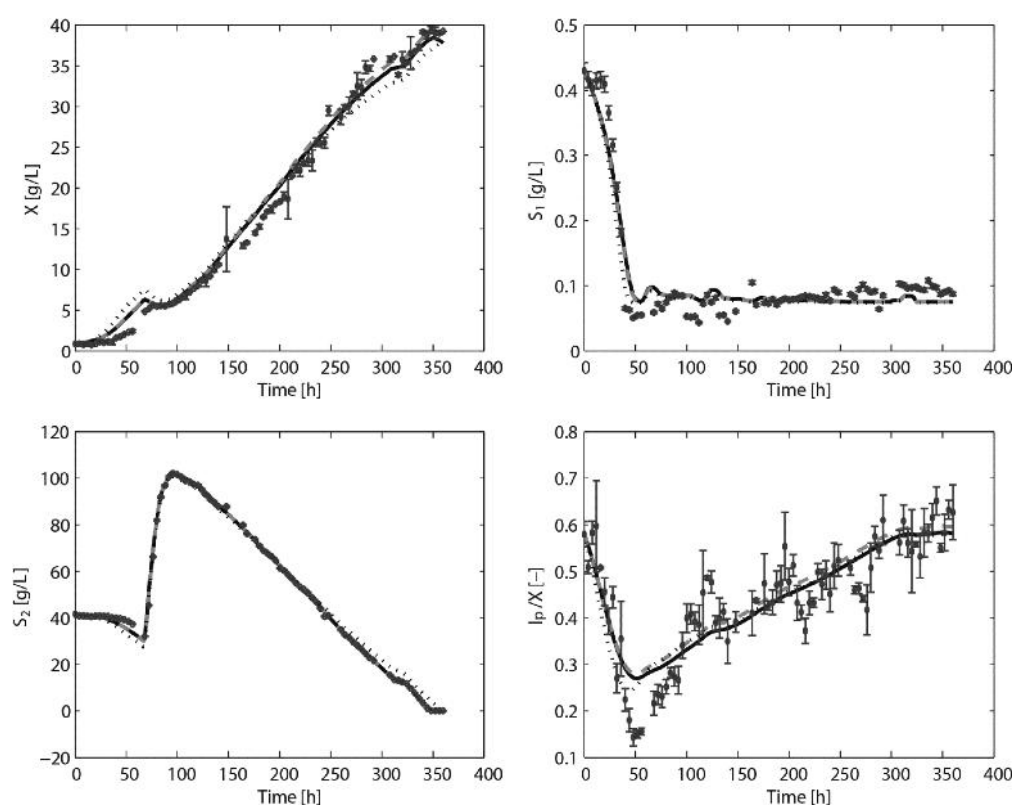


Figure 2. Fitting performance for the MFB model. Optimal 4-parameter model (continuous line), 12-parameter model achieved with eSS (segmented line), 12-parameter model proposed by Surisetty et al.¹⁷ (dotted line), and experimental data (circles) (the error bars represent the standard deviations of the measurements).

Table 3 shows the fitted and fixed values for our solution, the one proposed by Sacher et al.,¹ and the one achieved by SSm with the whole set of 14 fitting parameters.

Figure 3 shows the calibration accomplished with the iterative procedure (6-parameter model), the original 14-parameter model achieved with SSm, and the previous reparametrization achieved by Sacher et al. (a 9-parameter model).¹ It can be seen

Table 2. Assessment of Fitting Performance for the MFB Model (F_1 , Objective Function, Defined in Equation 1)

solution	F_1
Surisetty et al. (2010) ¹⁷ model	2.45×10^4
eSS 12-parameter model	9.55×10^3
optimal 4-parameter model	9.46×10^3

Table 3. Estimated and Fixed Parameters for the SSF Model^a

Parameters	Sacher et al. (2011) model			SSm 14-parameter model			Optimal 6-parameter model		
	Value	CI	CC	Value	CI	CC	Value	CI	CC
μ_m [1/h]	2.30E-01	-	-	3.24E-01	$\pm 2.32E-01$	1.43E+00	3.54E-01	$\pm 5.62E-03$	3.18E-02
k_d [g/g]	1.94E-02	$\pm 1.91E-03$	1.97E-01	4.60E-02	$\pm 5.94E-03$	2.58E-01	4.57E-02	$\pm 5.74E-04$	2.51E-02
k_U [g/mL]	4.87E-04	$\pm 4.85E-06$	1.99E-02	1.00E-03	$\pm 6.65E-04$	1.33E+00	5.26E-04	$\pm 6.81E-06$	2.59E-02
m_{CO_2} [1/h]	1.27E-01	$\pm 2.27E-02$	3.56E-01	1.88E-01	$\pm 4.00E-02$	4.24E-01	1.88E-01	$\pm 2.34E-03$	2.49E-02
m_{O_2} [g/mL]	5.42E-02	$\pm 9.75E-03$	3.60E-01	8.93E-02	$\pm 1.90E-02$	4.25E-01	8.93E-02	-	-
k_p [g/mL]	0E+00	-	-	9.44E-05	$\pm 8.63E-04$	1.83E+01	9.44E-05	-	-
Y_{NX} [g/g]	1.41E+01	$\pm 2.44E+00$	3.47E-01	1.59E+01	$\pm 3.17E-01$	3.98E-02	1.57E+01	$\pm 1.63E-01$	2.07E-02
k_N [1/h]	1.02E-04	$\pm 3.61E-05$	7.08E-01	2.81E-03	$\pm 2.42E-03$	1.72E+00	2.81E-03	-	-
Y_{CO_2X} [g/g]	1.96E+00	-	-	2.22E+00	$\pm 2.70E+00$	2.43E+00	2.22E+00	-	-
Y_{O_2X} [1/h]	3.80E+00	-	-	1.83E+01	$\pm 8.64E+01$	9.44E+00	1.83E+01	-	-
β_m [g/mL]	2.25E-04	$\pm 9.08E-05$	8.08E-01	6.05E-04	$\pm 2.17E-04$	7.15E-01	5.65E-04	$\pm 1.34E-05$	4.72E-02
k_i [g/g]	1.78E+05	$\pm 5.34E+06$	6.00E+01	4.16E+03	$\pm 2.02E+03$	9.72E-01	4.16E+04	-	-
Y_{SX} [g/g]	1.83E-01	-	-	1.70E-01	$\pm 1.32E-02$	1.56E-01	1.70E-01	-	-
m_S [g/g·h]	5.23E-02	$\pm 4.40E-02$	1.68E+00	5.00E-02	$\pm 1.96E-02$	7.83E-01	5.00E-02	-	-

^aSacher model is from ref 1. The estimation accuracy and significance are shown through 95% confidence intervals (CI) and coefficient of confidence (CC), respectively. The fixed parameters appear in black shaded cells. Note: 2.30E-01 notation represents, for example, 2.30×10^{-1} .

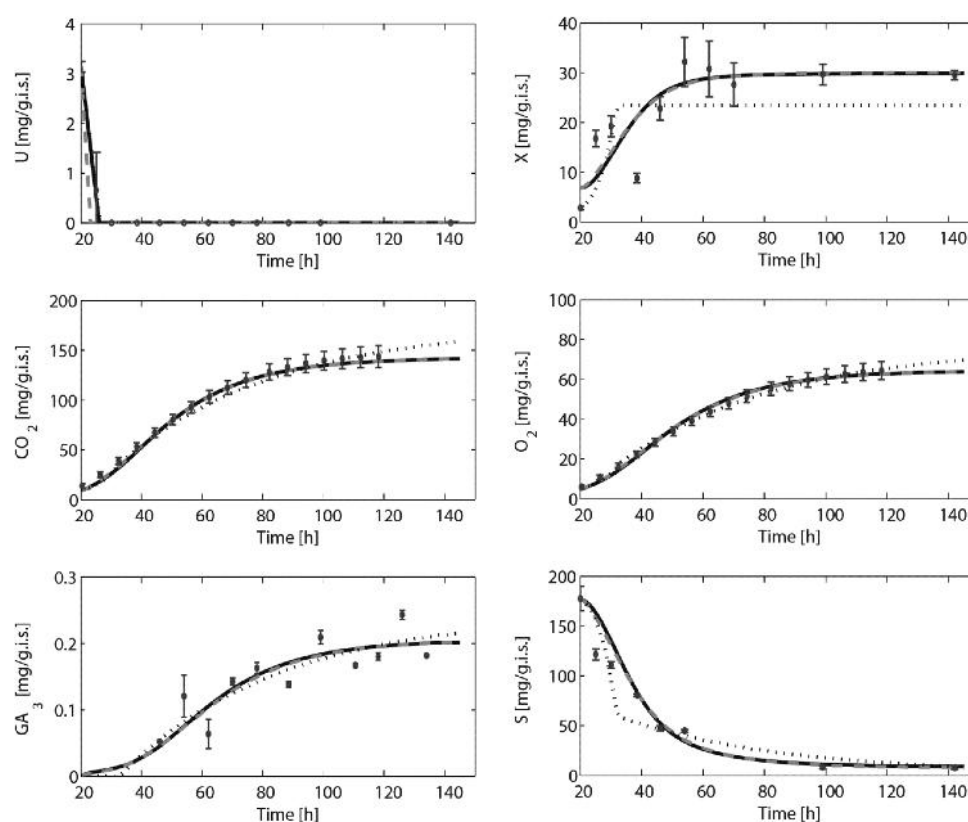


Figure 3. Fitting performance for the SSF model. Optimal 6-parameter model (continuous line), 14-parameter model achieved with SSm (segmented line), 9-parameter model proposed by Sacher et al.¹ (dotted line), and experimental data (circles) (the error bars represent the standard deviations of the measurements).

that our 6-parameter solution had a better fitting than the 14-parameter model for urea (U) and a significantly better fitting than the 9-parameter model for biomass (X). Furthermore, Table 4 shows that the objective function (F_2 , eq 2) for the

mentioned fittings is lower for our 6-parameter solution than for the other two models. Overall, with our reparametrization method we were able to obtain much better fits for the MFB and SSF models.

Table 4. Assessment of Fitting Performance for the SSF Model (F_2 , Objective Function, Defined in Equation 2)

solution	F_2
Sacher et al. (2011) ¹ model	1.137
SSm 14-parameter model	0.164
optimal 6-parameter model	0.156

Identifiability Analysis. *Microalgal Fed-Batch Bioreactor Model.* Identifiability analysis to the MFB model found several cases of high correlation among parameters reported by Surisetty et al.;¹⁷ μ_m with $q_{\min}$ and π_m , and ρ_m with S_0 , Y_{SX} , and k_π (in all possible combinations), as shown in Table 5a. For instance, ρ_m and k_π have a 100% positive correlation; this means that when parameters are estimated, an increase in both ρ_m and k_π (or a decrease of both) will yield the same numerical value for the objective function, so accurate estimates of both parameters are not feasible: the only option is to fix one of them and estimate the other. The same analysis, applied to the 12-parameter model fitted with the eSS algorithm, found high correlation values between μ_m and ρ_m , and between k_q , k_s , Y_{SX} , and k_m in various combinations (data not shown).

It is clear that a 12-parameter model is overparametrized; however, it is not evident which parameters should be fixed in order to have an identifiable model for there are several combinations of parameters set which could be explored. Our methodology provides a systematic procedure for obtaining a reduced set of parameters: the 4-parameter model obtained with HIPPO presented no excessive correlation among its fitting parameters (Table 5b).

Solid Substrate Fermentation Model. For the SSF model, the calibration with 9 parameters proposed by Sacher et al.¹ presented a correlation of -0.99 between k_U and k_N (Table 6a),

Table 6. Correlations between Estimated Parameters in the SSF Model for (a) the Original Calibration by Sacher et al.¹ and (b) the 6-Parameter Calibration Achieved by Our Procedure^a

a)

	k_d	k_U	m_{CO_2}	m_{O_2}	Y_{NX}	k_N	β_{glam}	k_i	m_S
k_d									
k_U	-0.16								
m_{CO_2}	-0.82	0.16							
m_{O_2}	-0.29	0.06	-0.11						
Y_{NX}	-0.81	0.18	0.56	0.24					
k_N	0.15	-0.99	-0.16	-0.06	-0.17				
β_m	0.09	-0.01	-0.11	-0.11	-0.01	0.01			
k_i	-0.10	0.01	0.12	0.12	0.01	-0.01	-0.91		
m_S	-0.47	0.10	0.11	0.11	0.83	-0.09	0.11	-0.12	

b)

	μ_m	k_q	k_U	m_{CO_2}	Y_{NX}	β_{glam}
μ_m						
k_d	-0.54					
k_U	0.90	-0.35				
m_{CO_2}	0.36	-0.83	0.25			
Y_{NX}	0.57	-0.81	0.31	0.51		
β_m	0.00	0.10	0.00	-0.11	0.00	

^aThe black shaded cell indicates correlated parameters.

which means that an increase in k_U and a decrease in k_N (or vice versa) will yield the same numerical result in the objective function as that described previously for the MFB model. Also the original calibration of 14 parameters presented several high correlations: k_N with μ_m , Y_{CO_2X} with m_{CO_2} , Y_{O_2X} with m_{O_2} , Y_{CO_2X} with Y_{SX} , and k_i with β_m (data not shown). On the other hand, the

Table 5. Correlations between Estimated Parameters in the MFB Model for (a) the Original Calibration by Surisetty et al.¹⁷ and (b) the 4-Parameter Calibration Achieved by Our Procedure^a

a)

	μ_m	$q_{\min}$	k_q	ρ_m	k_s	S_0	Y_{SX}	k_m	Y_{SP}	π_m	k_π	Y_{qX}
μ_m												
$q_{\min}$	0.99											
k_q	0.83	0.79										
ρ_m	-0.53	-0.61	-0.23									
k_s	0.12	0.20	-0.23	-0.74								
S_0	0.57	0.64	0.28	-0.99	0.66							
Y_{SX}	-0.81	-0.85	-0.44	0.79	-0.50	-0.79						
k_m	-0.86	-0.89	-0.69	0.72	-0.28	-0.75	0.84					
Y_{SP}	0.67	0.73	0.39	-0.98	0.66	0.98	-0.84	-0.81				
π_m	-0.99	-1.00	-0.80	0.61	-0.21	-0.64	0.84	0.88	-0.74			
k_π	-0.56	-0.63	-0.26	1.00	-0.72	-0.99	0.80	0.76	-0.98	0.63		
Y_{qX}	0.82	0.81	0.73	-0.40	-0.04	0.44	-0.69	-0.91	0.51	-0.79	-0.45	

b)

	μ_m	ρ_m	k_m	π_m
μ_m				
ρ_m	0.22			
k_m	-0.05	0.57		
π_m	-0.25	0.66	0.87	

^aBlack shaded cells denote correlated parameters.

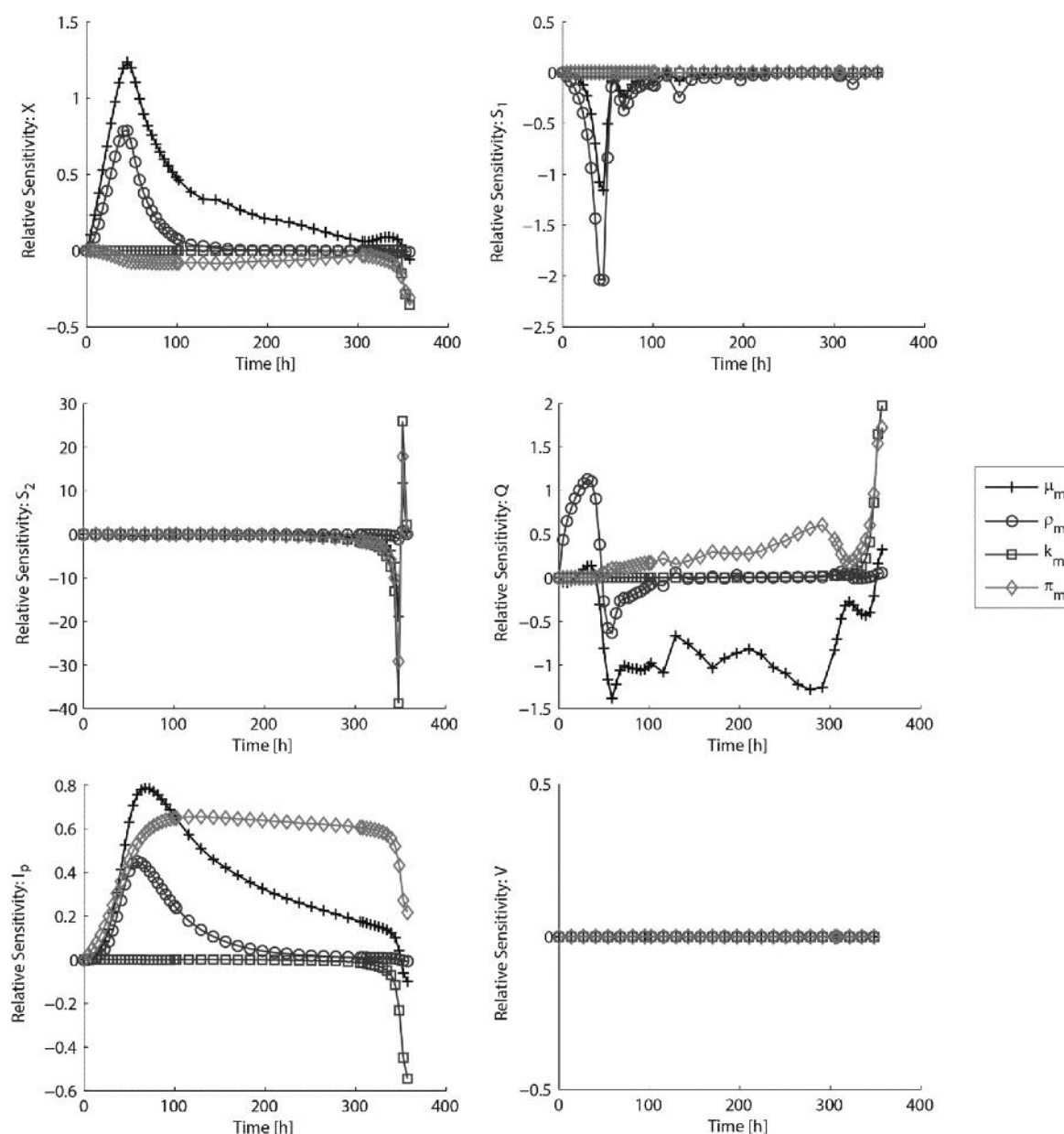


Figure 4. Temporal evolution of parameter sensitivities of the optimal 4-parameter solution for the MFB model.

achieved reparametrization of only 6 parameters did not show any correlation over 0.95 in absolute value (Table 6b).

Sensitivity Analysis. In this section we present results of sensitivity analyses performed upon the newly parametrized models found. We compared these results with those obtained with the originally fitted models.

Microalgal Fed-Batch Bioreactor Model. For the 4-parameter model, all fitting parameters highly influence at least one state variable, an observation that does not occur in the original 12-parameter solution proposed by Surisetty et al.¹⁷ (in which q_m , k_s , and k_π had very little impact on the model) nor in the eSS 12-parameter model found (in which k_π has almost no influence in the model). On the other hand, the insensitivity of the 8 fixed parameters was observed by replacing their values by those obtained by Surisetty et al.¹⁷ Figure S2 in the Supporting Information shows the fitting performance achieved by this option. Simple inspection shows that there is not a significant difference in the fitting performance with one set of parameter

values or the other one, which supports the idea that the exact values of these parameters are not relevant in the model output; i.e., they are insensitive.

Figure 4 shows temporal influence of estimated parameters in each state variable for the 4-model parameter. The active biomass concentration (X) is influenced by all fitting parameters, although with different patterns. μ_m and ρ_m strongly affect X through the first 100 h of the cultivation (upon nitrogen source starvation), whereas k_m and π_m have some influence in X only at the end of the cultivation (upon carbon source starvation). Nitrogen concentration (S_1) is just influenced by μ_m and ρ_m ; carbon concentration (S_2) is largely affected by all fitting parameters at the end of the cultivation, where S_2 is limiting. Furthermore, all 4 parameters affect the total nitrogen cellular quota (Q) and the total oil stored in cells (I_p) in a similar way: ρ_m during the first 100 h, k_m during the last 50 h, and μ_m and π_m during almost the whole cultivation. Finally, as expected, no parameter influences volume (V), because it is predetermined by fixed inputs and

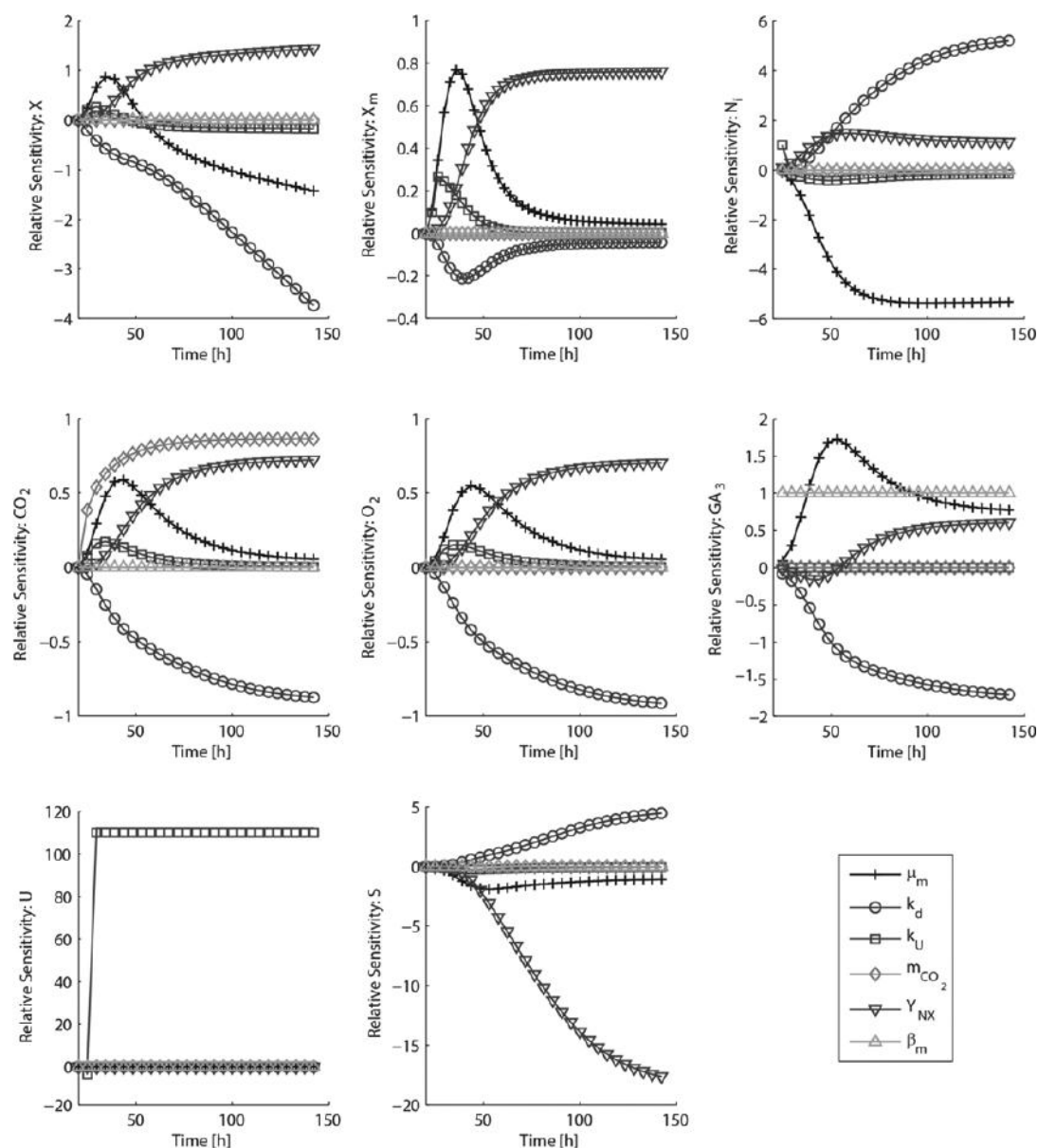


Figure 5. Temporal evolution of parameter sensitivities of the optimal 6-parameter solution for the SSF model.

outputs. These results are useful to design further experiments to improve the accuracy of estimated parameters. Since most sensitivities show a changing trend at 50 h (nitrogen limitation) and at 350 h (carbon limitation) of cultivation, new experiments should consider measuring more frequently around those times.

Solid Substrate Fermentation Model. In the final 6-parameter model obtained in the SSF case, all parameters influenced at least one state variable. This was not the case for the original 14-parameter model, in which k_p had a null influence in the model, and Y_{CO_2X} and Y_{O_2X} had close to a null influence. The solution of Sacher et al.¹ also had a parameter (k_i) with sensitivity close to zero. Figure S3 in the Supporting Information assesses the impact on the fitting performance of the 6-parameter model when the values of the 8 fixed parameters are changed, in this case with the values obtained by Sacher et al.¹ The calibrations shown are almost the same, confirming the insensitiveness of the chosen fixed parameters.

Figure 5 shows temporal sensitivities of each parameter for the final SFF model presented. Total biomass (X) is mostly influenced by 3 of the 6 parameters, with μ_m being most relevant

during the first 100 h of fermentation. k_d shows a strong negative influence in total biomass. This pattern is similar for measured biomass (X_m), except that in this case the impact of k_d is smaller. The intermediate nitrogen (N_I) is also affected by μ_m and k_d in an analogous pattern. In addition, parameter sensitivity evolutions of both CO_2 and O_2 behave similarly, with μ_m and k_U relevant at the beginning, and k_d and Y_{NX} at the end. The only difference is the noninfluence of m_{CO_2} in O_2 consumption. Gibberellic acid (GA_3) production depends on biomass parameters as well; μ_m influences the variable primarily at the beginning and k_d at the end of fermentation. GA_3 production is also sensitive to nitrogen uptake parameters (Y_{NX} and β_m), with β_m influencing GA_3 during all fermentation. Urea (U) equation is a special case since it follows a zero-order kinetics and has a discontinuity upon limitation. Therefore, temporal sensitivity reaches a high negative value in a short time, and afterward it becomes constant (urea-limited period). Finally, starch (S) consumption is mostly influenced by Y_{NX} at late stages of the process. In conclusion, more frequent measurements should be performed at the

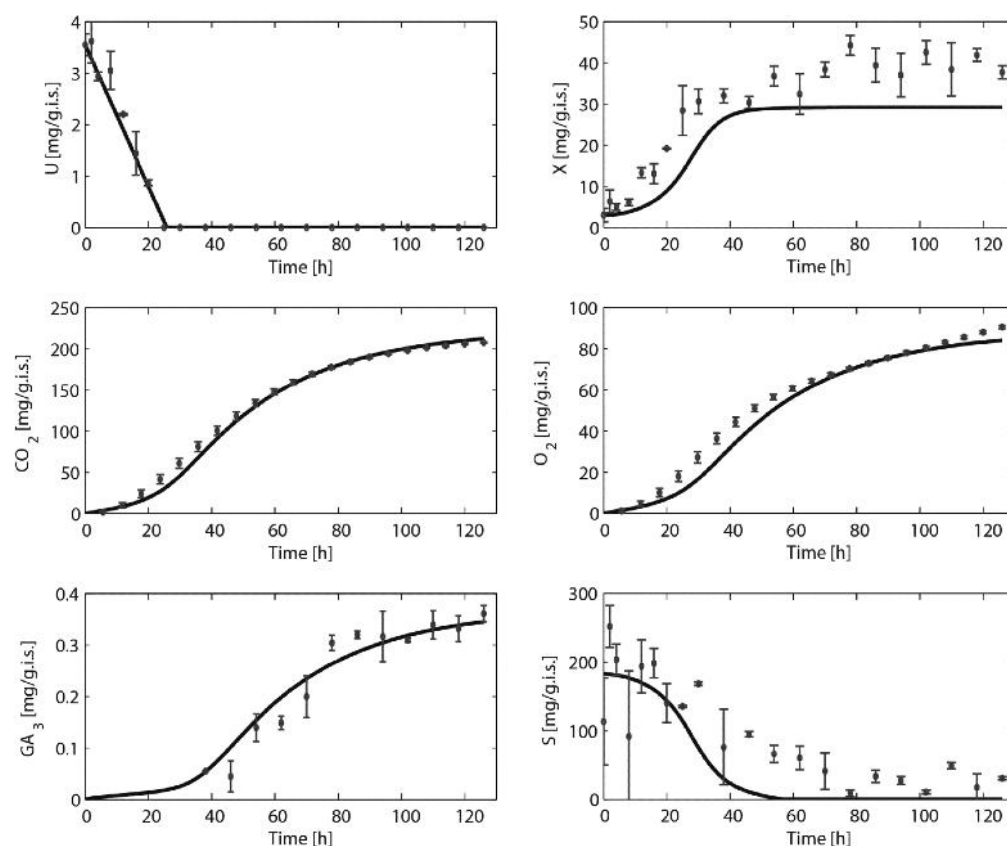


Figure 6. Fitting performance of the SSF model reparametrization in different experimental conditions. Reparametrized 6-parameter model (continuous line) and experimental data (circles) (the error bars represent the standard deviations of the measurements).

beginning (upon urea starvation) and at the end (upon starch starvation) of fermentation, because those periods are more critical in terms of parameter sensitivities.

Statistical Significance. Microalgal Fed-Batch Bioreactor Model. Table 1 summarizes the output of the statistical analysis done with the following models: optimal 4-parameter model yielded by our iterative procedure, full 12-parameter model identified with eSS, and 12-parameter model discussed in Surisetty et al.¹⁷ All estimates of the optimal 4-parameter model were significantly different from zero (CCs < 2) and accurate (smaller CIs than the other two models); the top value happens for k_m (CC = 1.43×10^{-4}). The CIs and CCs for the 12-parameter model identified with eSS were also small, although on average they were larger than the values obtained with the 4-parameter model. The top value occurred for $q_{\min}$ with CC = 0.04. Finally, the calibration found in Surisetty et al.¹⁷ had the largest CCs and CIs, reaching a value CC = 0.06 in the case of S_0 . Although in these three models all estimated parameters are significant, the present approach reduces the CI and CC values, which is a clear improvement for any model.

Solid Substrate Fermentation Model. The CI and CC values of the SSF model are shown in Table 3. The result of the original calibration by Sacher et al.¹ with 9 parameters has the largest CIs and CCs, with a top value of CC = 60 for k_r . The fitting achieved with the complete 14-parameter model is not good either: the worst estimates are Y_{O_2X} with a CC = 9.44 and Y_{CO_2X} with a CC = 2.43. For the 6-parameter calibration the worst estimate is β_m with CC = 0.0472; hence this is the only model—from the three discussed here—that has all of its parameters accurately estimated.

Methodology Validation. As a proof of concept, the proposed methodology was validated showing that the SSF 6-parameter model adequately calibrates different experimental conditions (the methodology was not validated for the MFB model because no additional experimental conditions were available). Using another data set of experimental conditions (temperature of 25 °C and water activity of 0.992),¹ we used the SSF 6-parameter model previously attained for calibration (i.e., we fixed m_{O_2} , k_p , k_N , Y_{CO_2X} , Y_{O_2X} , k_d , Y_{SX} , and m_S at the same values indicated in Table 3, and only estimated μ_m , k_d , k_U , m_{CO_2} , Y_{NX} , and β_m). The new estimated parameter values are shown in Supporting Information Table S3. We obtained satisfactory calibrations (Figure 6), indicating that our procedure accomplishes a model with identifiable, sensitive, and significant parameters, without losing predictive power.

Approach Limitations. Our proposed method differs from Balsa-Canto et al.² in that symbolic mathematical manipulations—such as deriving series expansion coefficients and constructing the corresponding identifiability tableaux—are not used; instead, parameter sensitivities are computed at each step of the method after a global optimization has been carried out, as shown earlier. Thus, we avoid lengthy symbolic manipulations to get definite answers on global structural identifiability, which is the first step of the Balsa-Canto et al.² methodology. These researchers have compared several available methodologies for assessing structural identifiability; they showed that those methodologies' performance depend strongly upon the process' mathematical equations and that only a generating series approach combined with identifiability tableaux seems the best choice.²⁹ Thus, our approach can be regarded as selecting the most relevant parameters to be globally estimated using numerical computations

based on real process data analysis, circumventing the aforementioned symbolic manipulation task.

The methodology presented here is heuristic, therefore, no mathematical proofs of global convergence of parameter estimates are given; only specific applications — carried out with actual experimental data — are shown to give a “proof of concept” of our methodology. Hence, our claim is that the present methodology is applicable for unstructured, dynamic bioprocess models and we can only conjecture that it might work as well for other dynamic bioprocess models.

Finally, we do not intend to specify further experimental designs to reestimate parameters fixed in the aforementioned process; we are only focused in properly fitting dynamic bioprocess models with already available data. In order to construct further experimental designs we would have to resolve the issue of whether the fixed parameters found are structurally unidentifiable or identifiable but poorly estimated.⁹ This issue is beyond the scope of the present work.

CONCLUSION

In this research we have proposed HIPPO: a novel heuristic iterative procedure, free of symbolic manipulations, which allowed a systematic reduction of the set of parameters to estimate in unstructured complex bioreactor models. The proposed method included a global scatter search optimization, avoiding local search limitations, and standard local regression diagnostics such as parameter identifiability, sensitivity, and significance analysis. An important advantage of this procedure is that a reduced set of all possible parameter combinations was explored in order to calibrate the model, which reduced the computational burden and improved the estimation solution. For example, for the MFB model with 12 fitting parameters, an exhaustive search would require 4,095 parameter estimations, while the proposed procedure required only 479 parameter estimations. For the SSF model with 14 fitting parameters, an exhaustive search would require 16,383 parameter estimations, of which our procedure only had to examine 918. Also, our methodology ensured that the proposed reparametrization only includes parameters with no identifiability, sensitivity, and significance problems.

The application of our method led to 110 models in the MFB case and 116 models in the SSF case; all of them had uncorrelated, sensitive, and significant estimated parameters. These are desirable characteristics in any modeling process. In addition, the methodology included analysis tools to discern which of the alternative parameter combinations better reproduces experimental data with fewer fitting parameters. Overall, the best options were a 4-parameter model for the MFB case and a 6-parameter model for the SSF case. Furthermore, the latter model was able to fit a different experimental condition, showing that it kept its predictive power.

The iterative procedure proposed in this work is intuitive, systematic, and easy to code. Hence, it can be used to develop mathematical models in other engineering areas as well.

ASSOCIATED CONTENT

Supporting Information

Figures S1–S3 showing the number of iterations as a function of the number of fixed parameters in comparison to the total possible combinations and model MFB and model SSF fitting performances, Tables S1–S7 listing nonproblem solutions from the MFB and SSF models, estimated parameters in the SSF model validation, model parameters for MFB and SSF models, and experimental data for MFB and SSF model calibrations, and

text detailing information about the fed-batch microalgae bioreactor model and solid substrate fermentation model. This material is available free of charge via the Internet at <http://pubs.acs.org>.

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Notes

The authors declare no competing financial interest.

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ABBREVIATIONS

CC = confidence coefficient
CI = confidence interval
FIM = Fisher information matrix
MFB = microalgal fed-batch bioreactor
SSF = solid substrate fermentation

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